

Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

Henrique J. S. Santos, Luís Gomes, Rafael Silva
Instituto Superior de Engenharia do Porto (ISEP)
Porto, Portugal
1180934@isep.ipp.pt

Abstract—Retail investors face a continuous flood of price movements and financial news but lack the interpretive tools available to institutions. We present CLARION, an explainable (XAI-first) financial-alert system for the US equity market that raises Telegram alerts from two independent triggers — abrupt price movements, detected as statistical anomalies, and incoming news — and attaches to every alert a fully traceable explanation. The core is a news–market correlation engine that retrieves analogous historical news with sentence embeddings and measures the impact those precedents had, in the manner of an event study and without lookahead. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components in a functional, reproducible system. On real data, semantic retrieval reaches a cross-ticker precision@5 of 0.514 ± 0.015 against 0.240 for random and 0.346 for a lexical baseline, and a volatility-normalised detector fires far more consistently across instruments (firing-rate range 0.015) than a fixed threshold (0.344). All results are reproducible from versioned scripts, with limitations stated explicitly.

Index Terms—explainable AI, retail investing, anomaly detection, sentence embeddings, information retrieval, event study, financial NLP

I. INTRODUCTION

Most Americans now own stock, and a growing share trade through commission-free mobile apps rather than through professional institutions [1]; the US equity market reached US\$62.2 trillion at the end of 2024 [2]. Yet the non-professional, or *retail*, investor must contend with continuous price movements and a constant stream of news without the tools that banks and funds take for granted. Meanwhile, the AI now spreading through finance is largely opaque [3], which is a poor fit for someone who must act on an alert and live with the result.

This paper presents CLARION, an explainable financial-alert system that does *not* attempt to predict prices. Instead, for each abrupt move or relevant news item, it shows what it detected, how it reached that conclusion, the sources, and which past events resemble the current one, alongside what happened to prices afterwards. The contribution is one of Artificial Intelligence *Engineering*: not a new algorithm, but the disciplined integration, application and critical evaluation of existing components into a working, explainable and reproducible system, together with a documented methodology for relating news to market impact.

II. RELATED WORK

Anomaly detection. Financial anomaly detection is dominated by unsupervised methods because labelled anomalies are scarce [4]. A rolling z-score is a transparent statistical detector; more expressive alternatives such as Isolation Forest [5] or deep detectors [6] trade interpretability for capacity. The need to normalise by recent volatility reflects volatility clustering, formalised by the ARCH/GARCH family [7], [8]; we use a non-parametric rolling estimate for transparency.

Explainability and trust. Surveys map the field and its transparent-versus-post-hoc divide [3]; a influential position argues for models that are interpretable by design in high-stakes settings [9], rather than post-hoc explanations of black boxes [10], [11]. Human-factors work frames the goal as *appropriate reliance* [12], and recent evidence warns that explanations can induce over-reliance [13]; we therefore explain with verifiable evidence rather than persuasive narrative.

Text representation and retrieval. Financial text analysis has moved from lexicons to learned representations; Sentence-BERT [14], built on the Transformer [15], yields directly comparable sentence vectors. Retrieval and its evaluation belong to information retrieval: the vector-space model [16], lexical ranking such as BM25 [17], and precision@ k as the standard ranked-retrieval metric [18].

Event studies and case-based reasoning. The event-study design measures returns around an event [19], [20]; market efficiency [21] explains why we *measure* reactions rather than predict them. Using retrieved past cases as evidence is case-based reasoning [22]. The FNSPID dataset [23] time-aligns news with prices, enabling a news-to-impact knowledge base.

III. THE CLARION SYSTEM

CLARION is a pipeline of single-responsibility components linked by a common schema. Two data layers share that schema: a *historical* layer (a knowledge base of past news, each entry storing date, ticker, headline, sentence embedding and measured post-event impact) and a *live* layer (free price and news feeds). Two triggers converge on one explanation step before delivery via Telegram.

Market trigger. Let r_t be the log-return on day t and let μ_t, σ_t be the mean and standard deviation over a trailing window of w days ending strictly before t . The day is flagged

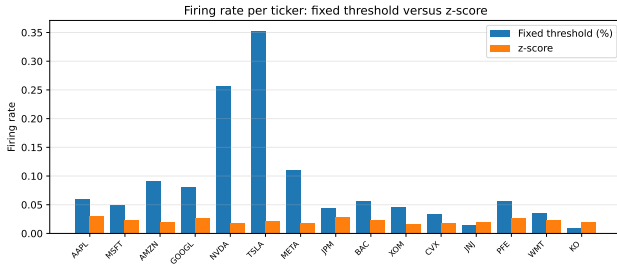


Fig. 1. Firing rate per ticker: a fixed percentage threshold varies dramatically with each stock’s volatility, while the volatility-normalised z -score is stable across all tickers.

when $|z_t| > k$ with $z_t = (r_t - \mu_t)/\sigma_t$. Every quantity behind the decision is exposed to the explanation engine, so the alert is faithful by construction.

News trigger. An incoming headline is embedded with Sentence-BERT and compared by cosine similarity against the knowledge base; the top- k precedents are returned with their observed impact at +1, +3 and +5 trading days, measured from the event-day close. The event day is the first trading day on or after the news date, and impact is accumulated strictly forward, which enforces a no-lookahead guarantee. This retrieve-and-reuse step is the case-based-reasoning core of the system.

Explanation. Each alert is rendered directly from the computed objects — the z -score with its window and threshold, or the retrieved precedents with their similarity scores and measured impact — and ends with an explicit no-prediction disclaimer.

IV. EVALUATION

The evaluation is fixed in advance. The anomaly detector is assessed primarily by the *consistency* of its firing rate across instruments of different volatility (a label-free argument) and secondarily against an extreme-move proxy. The correlation engine is assessed as an information-retrieval task by cross-ticker precision@ k against a same-sector relevance proxy, compared with random, recency and lexical baselines over five random seeds. All numbers below are reproduced by versioned scripts.

A. Anomaly detection

Over three years of daily data for fifteen large-cap US tickers, the fixed 3% threshold fires on $\sim 1\%$ of days for a calm stock but $\sim 35\%$ for a volatile one, whereas the z -score fires at a near-constant 2–3% for all. The range of firing rates across tickers is more than twenty times tighter for the z -score (0.015 vs 0.344; Fig. 1). Against the extreme-move proxy the z -score reaches $F_1 = 0.516$ versus 0.218 for the fixed rule, and a window ablation gives F_1 of 0.385, 0.516 and 0.678 for 10-, 20- and 60-day windows.

B. Precedent retrieval

On 3,714 real headlines across five sectors, semantic retrieval clearly outperforms every baseline (Table I). MiniLM

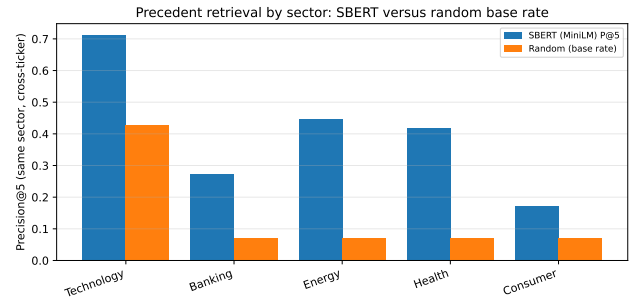


Fig. 2. Per-sector cross-ticker precision@5 against the random base rate. Retrieval beats the base rate in every sector; the lift is largest where the domain vocabulary is most distinctive.

TABLE I
CROSS-TICKER PRECISION@ k , MEAN $\pm$ STD OVER FIVE RUNS.

Method	P@5	P@10
SBERT (MiniLM)	0.514 $\pm$ 0.015	0.478 $\pm$ 0.011
SBERT (MPNet)	0.538 $\pm$ 0.011	0.506 $\pm$ 0.010
Lexical (hashing)	0.346 $\pm$ 0.011	0.314 $\pm$ 0.008
Recency	0.126 $\pm$ 0.006	0.106 $\pm$ 0.005
Random (base rate)	0.240 $\pm$ 0.004	0.240 $\pm$ 0.004

reaches a cross-ticker precision@5 of 0.514 (about $2.1\times$ the random base rate), the larger MPNet slightly more (0.538), and both far exceed the lexical baseline (0.346); recency is worst, below the random rate. Small standard deviations (~ 0.01) show the gap is robust. Per sector (Fig. 2), the lift over the base rate is largest where vocabulary is distinctive — energy (+0.377) and health (+0.348) — and smallest for the broad consumer sector (+0.100); technology has the highest raw precision (0.712) only because it dominates the corpus and so has a high base rate (0.429).

C. Faithfulness

Because each alert is rendered from the same objects the system computes, the explanation cannot diverge from the underlying computation; this is checked by an automated test that asserts the rendered alert reproduces each retrieved precedent’s date, ticker and measured impact.

V. DISCUSSION AND LIMITATIONS

The results support the design: transparent volatility-normalised detection is consistent where a fixed threshold is not, and semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines. The limitations are stated plainly. Relevance is a same-sector proxy, not a human judgement; the corpus is a recent window from a free service rather than the multi-year FNSPID history; impact is a raw post-event return, not a market-adjusted abnormal return; and no human usefulness study has yet been run. A further, conceptual limitation is that embedding similarity captures *theme* rather than *direction*: a retrieved cluster can mix positive and negative outcomes (a positive headline may retrieve a competitive-threat cluster), so the reported mean is

evidence about a theme, not a directional forecast, which is why precedents are always shown individually and every alert disclaims prediction. By design, the system performs no price prediction or trading, so profit-based metrics are out of scope.

VI. CONCLUSION

CLARION shows that useful financial alerting for retail investors can be both transparent and reproducible: a volatility-normalised detector that explains every flag, and a news-market correlation engine that retrieves analogous precedents and reports their measured impact as evidence rather than prediction. Future work is the full multi-year knowledge base, a human relevance and usefulness study, and market-adjusted impact measurement.

REFERENCES

- [1] Gallup, “What percentage of americans own stock?” 2025. [Online]. Available: <https://news.gallup.com/poll/266807/percentage-americans-owns-stock.aspx>
- [2] Securities Industry and Financial Markets Association (SIFMA), “2025 capital markets fact book,” 2025. [Online]. Available: <https://www.sifma.org/resources/research/fact-book/>
- [3] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bannetot, S. Tabik, A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, “Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI,” *Information Fusion*, vol. 58, pp. 82–115, 2020.
- [4] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Computing Surveys*, vol. 41, no. 3, pp. 15:1–15:58, 2009.
- [5] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.
- [6] G. Pang, C. Shen, L. Cao, and A. van den Hengel, “Deep learning for anomaly detection: A review,” *ACM Computing Surveys*, vol. 54, no. 2, pp. 38:1–38:38, 2021.
- [7] R. F. Engle, “Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation,” *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [8] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [9] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [10] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774.
- [11] M. T. Ribeiro, S. Singh, and C. Guestrin, ““why should I trust you?”: Explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.
- [12] J. D. Lee and K. A. See, “Trust in automation: Designing for appropriate reliance,” *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004.
- [13] G. Bansal, T. Wu, J. Zhou, R. Fok, B. Nushi, E. Kamar, M. T. Ribeiro, and D. Weld, “Does the whole exceed its parts? the effect of AI explanations on complementary team performance,” in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021.
- [14] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.
- [15] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2019.
- [16] G. Salton, A. Wong, and C. S. Yang, “A vector space model for automatic indexing,” *Communications of the ACM*, vol. 18, no. 11, pp. 613–620, 1975.
- [17] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 4, no. 1–2, pp. 1–174, 2009.
- [18] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, England: Cambridge University Press, 2008.
- [19] E. F. Fama, L. Fisher, M. C. Jensen, and R. Roll, “The adjustment of stock prices to new information,” *International Economic Review*, vol. 10, no. 1, pp. 1–21, 1969.
- [20] S. J. Brown and J. B. Warner, “Using daily stock returns: The case of event studies,” *Journal of Financial Economics*, vol. 14, no. 1, pp. 3–31, 1985.
- [21] E. F. Fama, “Efficient capital markets: A review of theory and empirical work,” *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [22] A. Aamodt and E. Plaza, “Case-based reasoning: Foundational issues, methodological variations, and system approaches,” *AI Communications*, vol. 7, no. 1, pp. 39–59, 1994.
- [23] Z. Dong, X. Fan, and Z. Peng, “FNSPID: A comprehensive financial news dataset in time series,” *arXiv preprint*, 2024.